$(-2)*(-1) = |(-2)-(-1)| - (-2)\times(-1)$

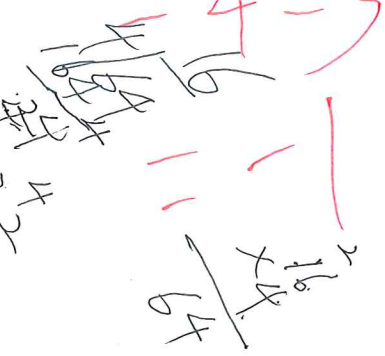
$= |-1| - 2 = 1 - 2 = -1$

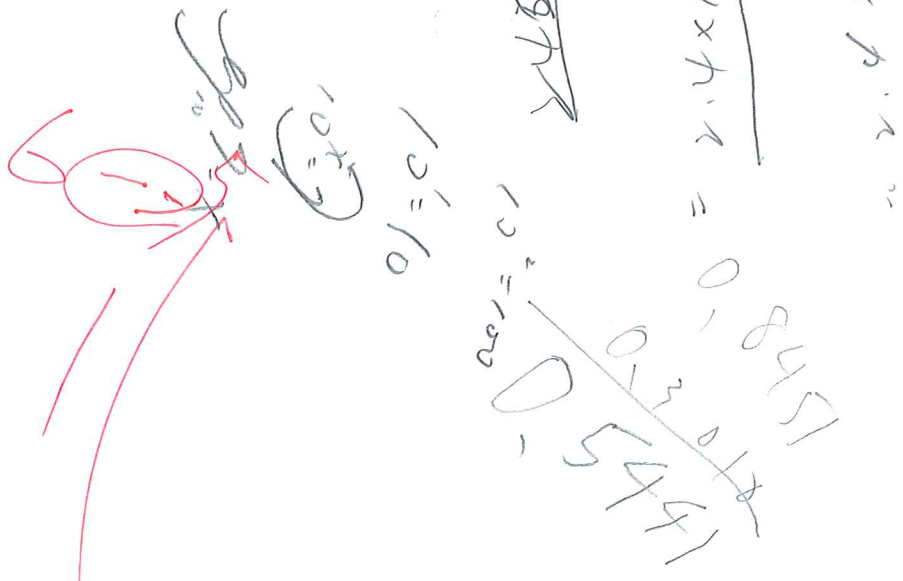
$5*(-1) =$

$= |(+5)-(-1)| - (+5)\times(-1)$

$= 4 - 5$

$= -1$





(5) B(-7)

中点の公式 $\frac{5+(-7)}{2} = -1$

距離 $5-(-7) = 12$

$-42 \times$

$\frac{1}{2}x^2 + x - 2 = \frac{1}{2}(x^2 + 2x - 4)$

$= \frac{1}{2}\{(x+1)^2 - 1 - 4\}$

$= \frac{1}{2}(x+1)^2 - \frac{5}{2}$

(2) $y = x^2 - 2x + 1 = (x-1)^2$

$= 11 \times (-3+1)$

$= -22$

$y = x^2 - 2x + 1$

$= (x-1)^2$

$= (-3-1)^2$

$= (-4)^2$

$= 16$

$y = x^2 - 2x + 1$

$= (-2)^2 + 4 + 1$

$= 4 + 4 + 1$

$= 9$

$y = x^2 - 2x + 1$

$= 5^2 - 10 + 1$

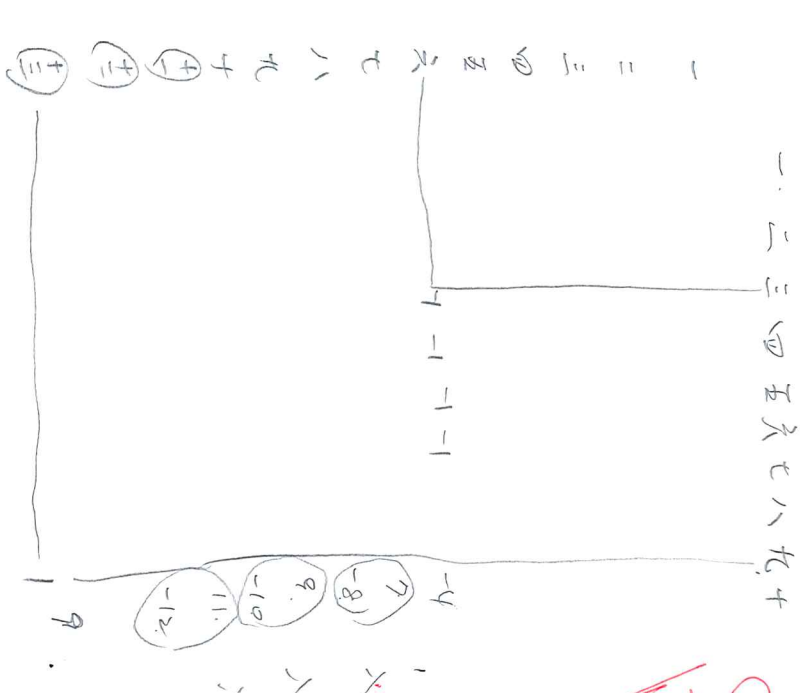
$= 25 - 10 + 1$

$= 16$

$42 \times 72 = 1884 + 2940 = 3024$

$16 \times 4 = 64$

$22 \times 22 = 484$



$(-18)+8\div(-4)+(-3)\times 4$

$=(-18)+(-2)+(-12)$

$=$

$(64+7\times(-8))\div 4$

$=(-9)\times 8$

$=-72$

10

64

56

8

$6\times[(-4)-(-6)]+(-4)$

$=6\times 2+(-4)$

$=12+(-4)$

8

56

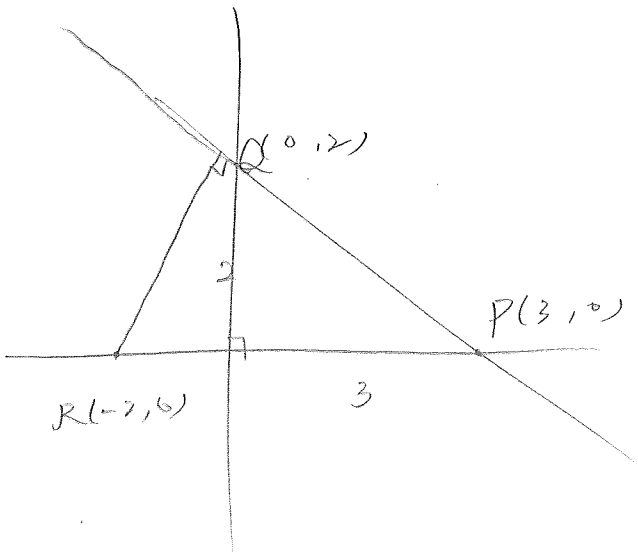
9

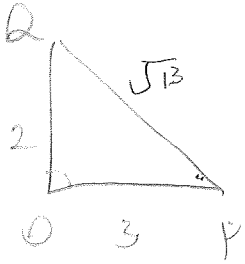
200 2 角 5

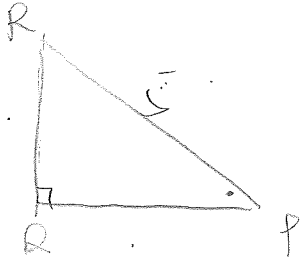
$x = 10000$

$(x-4)(x+5) - (x+4)(x-3)$

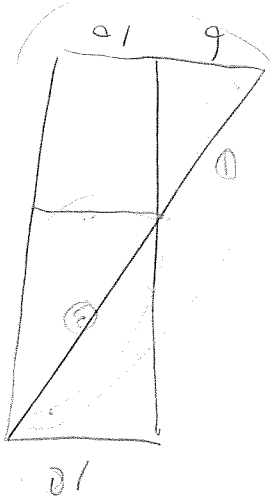
$(x^2+x-20) - (x^2+x-12)$





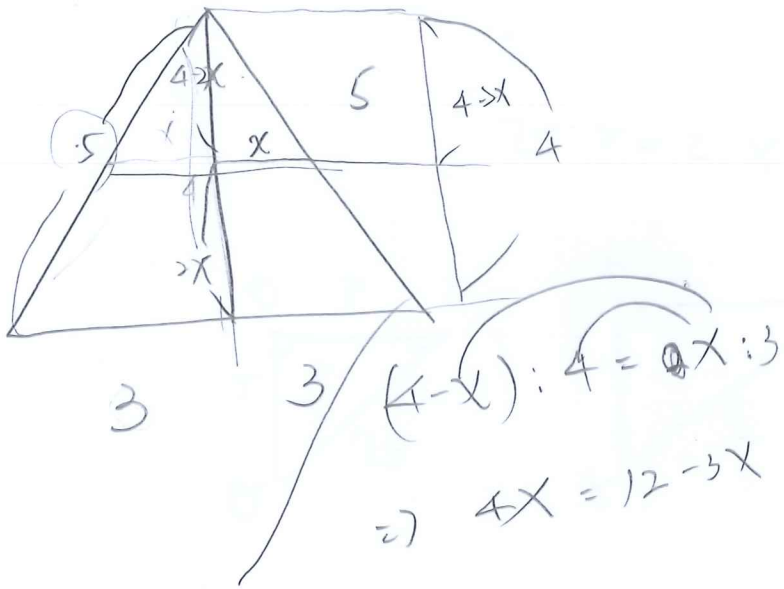


$\sqrt{13} : 2 = 5 : x$



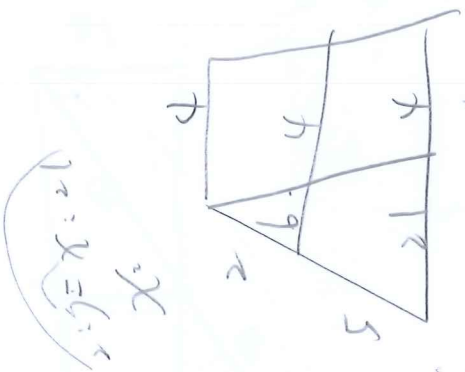


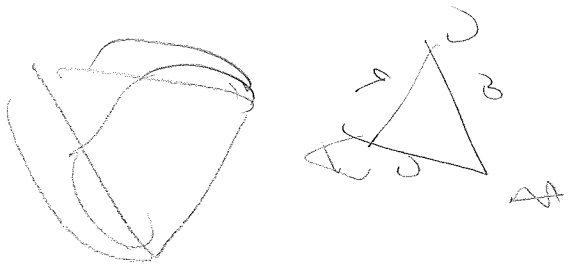
$\frac{9}{1} : \frac{5}{1} : \frac{4}{1} =$

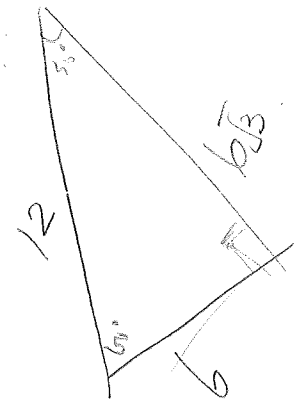


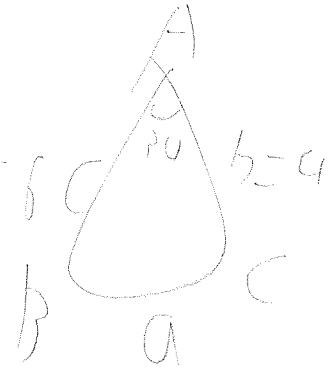
$(4-x):4=3x:3$

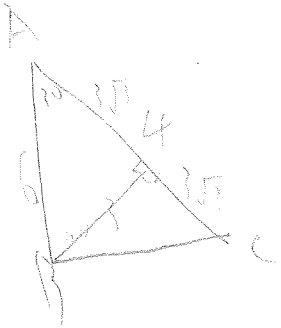
$\Rightarrow 4x = 12-3x$

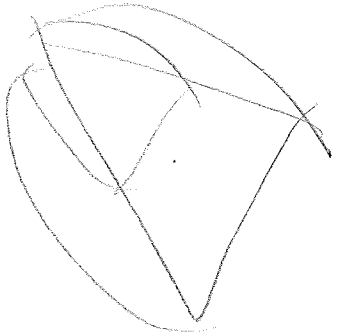

x
x
x

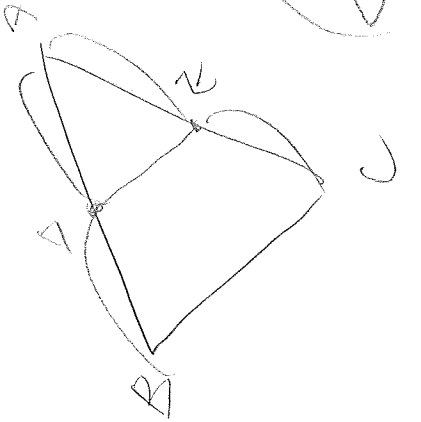


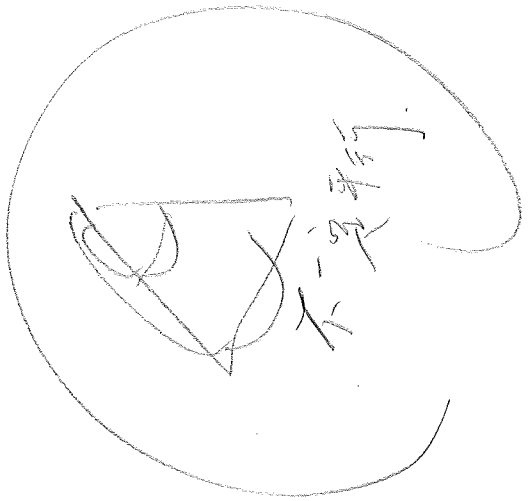




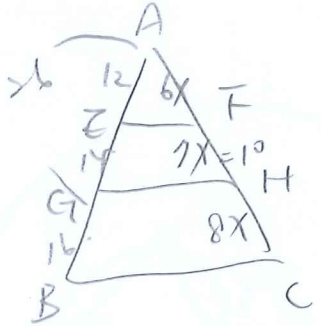




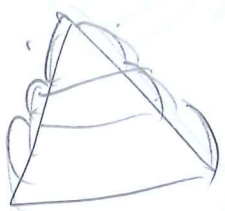




$\angle A : \angle B = 2 : 3$



$(3t+20) : (5t+20) : (6t-40) = 7 : 11 : 16$

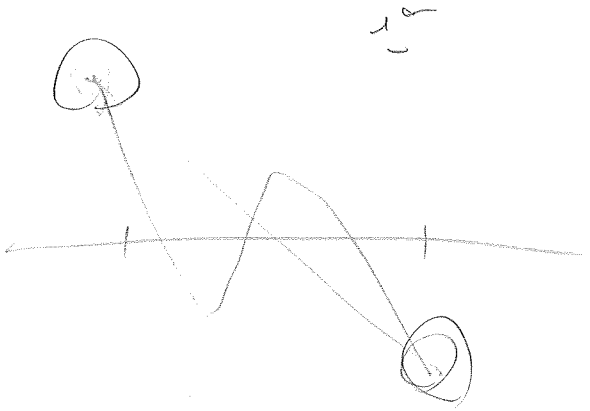


$(3t+20) : (5t+20) = 7 : 11$

$f(x) = x^3 - 2x^2 + ax - 1$

$\lim_{x \to 3^+} f(x) = \lim_{x \to 3^+} -2x + 11 = 5$

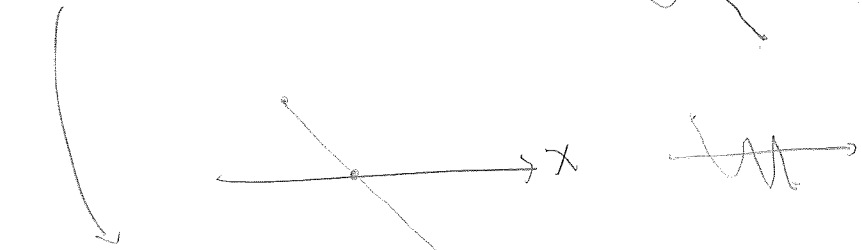
$\lim_{x \to 3^-} f(x) = \lim_{x \to 3^-} x^2 + ax + b = 9 + 3a + b$



$\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} x^2 + ax + b = b$

$\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} x - 2 = -2$

$f(0)f(1) < 0$



$(0 - 0 + 0 - 1)(1 - 2 + a - 1) < 0$

$(-1)(a - 2) < 0$

$a - 2 > 0$

$a > 2$

$\{x \mid x \in \mathbb{R},$

$$2^{2x+3y-2} = 2^{2x} \cdot 2^{3y} \cdot 2^{-2}$$

$$2^2 \times 2^1 = 2^3$$

$$\frac{\left(5^{x} + 5^{-x}\right) + \left(5^{x} - 5^{-x}\right)}{\left(5^{x} - 5^{-x}\right) \times 3}$$

$$\frac{\sqrt{7} - x^2 + 3}{x + 5}$$

$$\frac{x+5}{x^2+5}$$

$$\frac{x+5}{x^2+5}$$

$$\frac{x+5}{x^2+

解法： $\frac{|N-95|}{95} \times 100\% < 5\%$.

$\Rightarrow \frac{|N-95|}{95} < \frac{5}{100}$

$\Rightarrow |N-95| < 95 \times 0.05$

$\Rightarrow 95-4.75 < N < 95+4.75$

$\Rightarrow 90.25 < N < 99.75$

依題 N 可為 91, 92, 93, 94, ～, 99 共9個　Ans: (4)

2. 思緒：釣魚法 $\log_3 x = 6 \Leftrightarrow \log_3 6 = x$.

解法：

$100 e^{\frac{135}{100}} = 135$

$\Rightarrow e^{\frac{135}{100}} = \frac{135}{100}$

$\Rightarrow \ln e^{\frac{135}{100}} = \ln \frac{135}{100}$

$\Rightarrow n = \frac{100}{3} \ln e^{\frac{135}{100}}$

$\Rightarrow n = \frac{100}{3} \ln \frac{135}{100}$　Ans: (5)

3. 思緒：若 $AB = I$ 則 B 為 A 的反方陣.

AB 為方陣且

解法：依題意，$A\begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ 則可得 $A^{-1} = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$　Ans: (2)

4. 思緒：記得東經 0° 和西經 0° 同一位置，東經 180° 和西經 180° 同位置

解法：甲乙兩船所在的半徑比為 1:2，故所走的角度比為 2:1（甲乙兩船走走相同）.

甲走了 $(180-169)+(180-171) = 11+9 = 20$ 度，所以乙走了 10 度

故走到東經 $140+10 = 150$ 度.

Ans: (4)

解法：根据题意，

第一次	二	三	四	五	六	七
预定金额(元)	$\frac{3}{2}N$	$\frac{3^2}{2^2}N$	$\frac{3^3}{2^3}N$	$\frac{3^4}{2^4}N$	$\frac{3^5}{2^5}N$	$\frac{3^6}{2^6}N$
投注金额(元) N	$\frac{3}{2}N$	$\frac{3^2}{2^2}N$	$\frac{3^3}{2^3}N$	$\frac{3^4}{2^4}N$	$\frac{3^5}{2^5}N$	$\frac{3^6}{2^6}N$
		✓		✓		✓

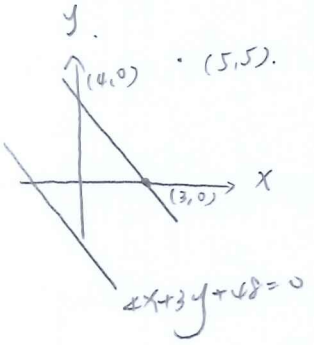
Ans: (3)

6. 思绪：若无法判断领海有无包含点 $(5,5)$ 可画图则较易解决

回忆平行线距离公式：若 $l_1: ax+by+k_1=0$，$M: ax+by+k_2=0$ 则 l_1, M 距离为 $\frac{|k_2-k_1|}{\sqrt{a^2+b^2}}$

解法：依题意，设另一直线为 $4x+3y+k=0$.

则 $\frac{|k-(-12)|}{\sqrt{4^2+3^2}}=12 \Rightarrow |k+12|=60 \Rightarrow k+12=\pm 60 \Rightarrow k=48$ 或 $k=-72$ (不合)



Ans: 1

7. 思绪：可用中文表相关条件概率定义记得则较易得解.

解法：$\frac{P}{Q}=\frac{P(\text{至少有两个福袋中奖}\mid A\text{中奖})}{P(A\text{中奖}\mid\text{至少有两福袋中奖})}=\frac{\frac{P(\text{至少有两个福袋中奖且 A中奖})}{P(A\text{中奖})}}{\frac{P(A\text{中奖且至少有两福袋中奖})}{P(\text{至少有两个福袋中奖})}}$

$=\frac{P(\text{至少有两福袋中奖})}{P(A\text{中奖})}=\frac{\frac{3}{4}\times\frac{2}{3}\times(1-\frac{1}{2})+\frac{3}{4}\times(1-\frac{2}{3})\times\frac{1}{2}+(1-\frac{3}{4})\times\frac{2}{3}\times\frac{1}{2}+\frac{3}{4}\times\frac{2}{3}\times\frac{1}{2}}{\frac{3}{4}}=\frac{17}{18}$

Ans: 12,

8. 思绪：第3解振幅、周期、平移（左加右减上加下减）与函数的关系

解法 $y=3\sin(\frac{\pi}{5}(x+5))+3$（3：振幅3；$+5$：左移5；$+3$：上移3）

周期为 $\frac{2\pi}{\frac{\pi}{5}}=10$.

Ans: (1)(5)

(2) 不同，振幅为6

(2) 不同，$y=3\sin(\frac{11}{5}\pi x+\pi)+3$，周期为 $\frac{2\pi}{\frac{11}{5}\pi}=\frac{10}{11}$

(3) 相同，$y=3\sin(\frac{\pi}{5}x-\pi)+3=3\sin(\frac{\pi}{5}(x-5))+3$，是向右移5，会与题目函数重合.

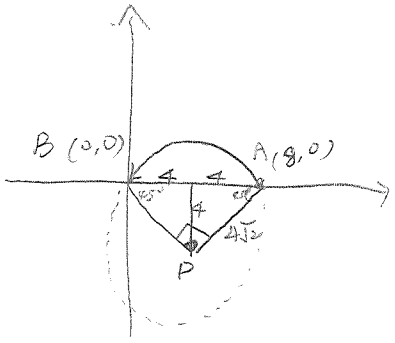
(4) 不同，$x=0$ 代入 $y=3\sin(\frac{\pi}{5}x+\pi)+3$ 得 $3\sin\pi+3=3$，但代入 $y=-3\sin(\frac{\pi}{5}x)-3$ 得 $-3\sin 0-3=-3$.

(5) 相同，如图所示  题目 

解法：依题意，(8,0)，(0,0)的中垂线为 $x=\frac{8+0}{2}\Rightarrow x=4$　　Ans：(1)

19. 思绪：求直线方程式要有斜率和点，要求圆的方程式要有圆心和半径

解法：依题意如下图 可知△ABP为等腰直角△ 故P(4,-4)



L的直线方程式为 $y=x-8$，P(4,-4)代入L $\Rightarrow -4=4-8$ (合)

故P在L上

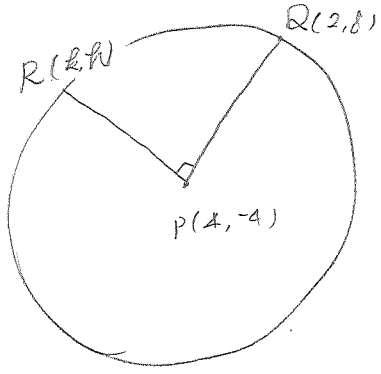
也可知甲圆轨迹方程式为 $(x-4)^2+(y+4)^2=(4\sqrt{2})^2$（所在的圆）

$\Rightarrow (x-4)^2+(y+4)^2=32$.

Ans：$(x-4)^2+(y+4)^2=32$

20. 思绪：向量垂直的积为零，若分不出哪个向量可画相对图看.

解法：依题意如下图 可令R(h,k)



因此 $\overrightarrow{PQ}=(-2,12)$，$\overrightarrow{PR}=(h-4,k+4)$.

又 $|\overrightarrow{PQ}|=|\overrightarrow{PR}|$ 且 $\overrightarrow{PQ}\perp\overrightarrow{PR}$ 则可得 $\overrightarrow{PR}=(12,2)$ 或 $(-12,-2)$（依合，是R……逆时P0°…）

所以 $\overrightarrow{PR}=(-12,-2)=(h-4,k+4)\Rightarrow h=-8,\ k=-6$

即 R(-8,-6).

Ans：$\overrightarrow{PR}=(-12,-2)$

R(-8,-6).

解法：可将下表填足

	有近	无	
有性	$p-\frac{1}{6}$	$\frac{1}{2}-p$	$\frac{1}{3}$
无	$\frac{2}{3}-p$	p	$\frac{2}{3}$
	$\frac{1}{2}$	$\frac{1}{2}$	1

由题意，有近视……有性可占少数

$p-\frac{1}{6}<\frac{2}{3}-p \Rightarrow p<\frac{5}{12}$

有性……无近……

$\frac{1}{2}-p<p-\frac{1}{6} \Rightarrow \frac{1}{3}<p$

综合上述 $\frac{1}{3}<p<\frac{5}{12}$

Ans. $\frac{1}{3}<p<\frac{5}{12}$.

16. 思绪：由两点可知一次函数，有三点可知二次函数，二次函数求极值可配完全平方.

解法：依题意 $f(x)=\frac{4-0}{5-1}(x-1)=x-1$.

可令 $g(x)=ax^2+bx+c$

$$\begin{cases} g(1)=0=a+b+c \\ g(5)=4=25a+5b+c \\ g(2)=2=4a+2b+c \end{cases} \Rightarrow a=-\frac{1}{3},\ b=3,\ c=-\frac{8}{3} \Rightarrow g(x)=-\frac{1}{3}x^2+3x-\frac{8}{3}.$$

则 $g(x)-f(x)=-\frac{1}{3}x^2+2x-\frac{5}{3}$

$=-\frac{1}{3}(x-3)^2+\frac{4}{3}$.

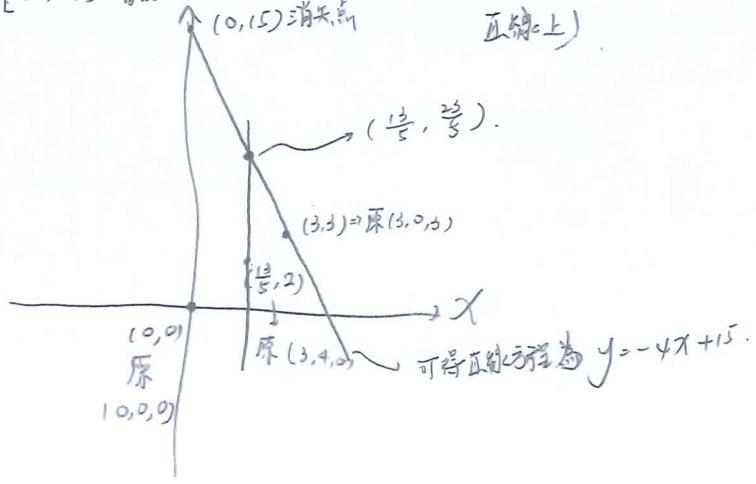
因此 $g(x)-f(x)$ 有最大值 $\frac{4}{3}$

Ans. $\frac{4}{3}$.

17. 思绪① 空间中与y轴平行的直线，在画面的消失点为(0,15)即空间中与y轴平行画在平面上这些直线都会过(0,15).

② 空间中与x轴平行的直线，在画布上都与y轴平行即. 空间中与x轴平行的直线画在平面上x的坐标点皆相同.

解法：依题意如下图 [(3,4,3)画在画布上] 画为 $(\frac{12}{5},2)$ 原(3,4,0) 有同样的x轴坐标，令为(3,3)原(3,0,3)和(0,15)在同条直线上)



Ans: $\frac{2}{3}$.

解法：(1) 錯誤．因為 $f(x)$ 為 $(1-x)(2-x)(4+x)$ 之倍式故餘式為 0．

(2) 正確，$f(x)=[-(x-2)-1][(x-2)^2-(x-2)+6]$

$=-(x-2)^4-7(x-2)^3-6(x-2)^2$ 故 $c=-6$．

(3) 錯誤，$f(x)>0 \Rightarrow -(x-1)(x-2)^2(x+4)>0$

$\Rightarrow (x-1)(x-2)^2(x+4)<0$



$\Rightarrow -4<x<1$．

(4) 正確；$\dfrac{f(2026)}{f(-2022)}=\dfrac{(-2025)\times 2024^2\times 2030}{2025\times 2024^2\times(-2018)}>1$．

(5) 錯誤，雖(4) $\dfrac{f(2026)}{f(-2022)}>1$，但 $f(-2022)<0 \Rightarrow f(2026)<f(-2022)$

10 思緒：① 標準差 = $\sqrt{\text{變異數}}$。標準差、變異數會用來描敘或呈現

② 過為直線，斜率 $m=r\cdot$ 坐標 $(\bar{x},\bar{y})$

解法：(1) 正確，B 的標準差 $\sigma_B=\frac{1}{2}\sigma_A=\frac{1}{2}\sqrt{225}=\frac{15}{2}$．

(2) 錯誤，無法得知

(3) 錯誤，B 濃度 $=\frac{1}{2}$ A 濃度 則 相關係數為 1

(4) 正確，身長平均數為 65 單位，A 濃度平均數為 50 單位 會在 過 $(\bar{x},\bar{y})$ 為直線上

(5) 錯誤，$m=r\cdot\dfrac{\sigma_y}{\sigma_x}=0.75\times\dfrac{\frac{15}{2}}{\sqrt{100}}=\dfrac{9}{16}$

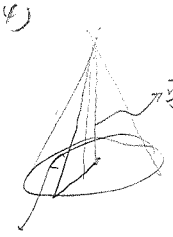
Ans：(1)(4)

11 思緒：圓：平面與軸線垂直　　軸與平面　軸與母線

橢圓：平面只切圓錐一側，且夾角大於母線夾角

拋物：平面與其中一條母線平行

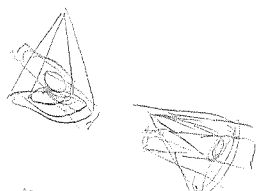
雙：平面與軸線平行，或夾角小於母線夾角，同時切到上下兩方圓錐



解法：(1) 錯誤：若大燈罩 Γ 是橢圓則 軸與平面夾角大於 軸與母線夾角

則小燈罩 與軸與平面夾角大於 軸與母線夾角 故 Γ 為橢圓

因此



(2) 正確：若大燈罩 Γ 是拋物線則 平面與其中一條母線平行 因此 小燈罩 軸與平面夾角大於 軸與母線夾角

故 Γ 為橢圓

(3) 錯誤：若大燈罩 Γ 是雙曲線則 平面與軸線平行 或夾角小於母線夾角，因此無法得知小燈罩的形狀

故 無法得知 Γ

(4) 錯誤：若小燈罩 Γ 是拋物線則 平面與其中一條母線平行 因此 大燈罩 Γ 平面與軸夾角小於母線夾角 故 Γ 為雙曲

(5) 正確：若小燈罩 Γ 是雙曲線則 平面與軸線平行或夾角小於母線夾角，因此 大燈罩 Γ 平面與軸 ⋯ 故 Γ 為雙曲

解法(1)正確，$a_1 = 100\times\frac{1}{2} + 50\times\frac{1}{2} = 75$.

(2)錯誤，不管幾次稀釋A瓶和B瓶糖含量總重為100公克

所以B瓶的含糖量為$100 - a_n$公克

(3)正確，$a_{n+1} = \frac{1}{2}a_n + \frac{1}{2}b_n = \frac{1}{2}a_n + \frac{1}{2}(100 - \frac{1}{2}a_n)$.

(4)正確，承(3) $a_{n+1} = \frac{1}{2}a_n + 50 - \frac{1}{4}a_n$

$\Rightarrow a_{n+1} = \frac{1}{4}a_n + 50$

$\Rightarrow (a_{n+1} - \frac{200}{3}) = \frac{1}{4}(a_n - \frac{200}{3})$

故可找到實數c滿足數列$\langle a_n - \frac{200}{3}\rangle$為公比是$\frac{1}{4}$的等比數列

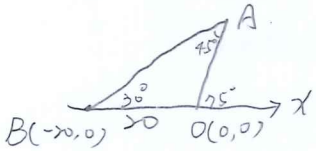
(5)錯誤，$(a_{100} - \frac{200}{3}) = (\frac{1}{4})^{99}(a_1 - \frac{200}{3})$

$\Rightarrow a_{100} = (\frac{1}{4})^{99}\times(75 - \frac{200}{3}) + \frac{200}{3} > \frac{200}{3} > 60$.

Ans: (1)(3)(4).

13. 思緒：正弦定理 $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$.

解法：



依題意如左圖則可算出$\angle BAO = 45°$

欲求$\overline{OA}$，可由正弦定理 $\frac{20}{\sin A} = \frac{\overline{OA}}{\sin B}$

$\Rightarrow \frac{20}{\sin 45°} = \frac{\overline{OA}}{\sin 30°}$

$\Rightarrow \frac{20}{\frac{1}{\sqrt{2}}} = \frac{\overline{OA}}{\frac{1}{2}}$

$\Rightarrow \overline{OA} = 10\sqrt{2} \approx 14.14$

Ans: (4)

14. 思緒：相鄰將其綁在一起視為一物

解法：依題意排法如下

(34) (56) (1)(2) 其中3.4可互換，5.6可互換

故所求為$4!\times 2!\times 2! = 96$

Ans: 96

$\cos\alpha + \cos\beta - \cos\alpha\cos\beta + \sin\alpha\sin\beta = \sqrt{2}$

$\sin\alpha\sin\beta + \cos\beta(1-\cos\alpha) = \sqrt{2} - \cos\alpha$

→ 求 $\cos\alpha$ 極值先視為常數

↓ 疊合

$\sqrt{\sin^2\alpha + (1-\cos\alpha)^2}\sin(\beta + ??) = \sqrt{2} - \cos\alpha$

$\sqrt{\sin^2\alpha + \cos^2\alpha + 1 - 2\cos\alpha}$

$=$

$\sqrt{2 - 2\cos\alpha}$

當 $\sin(\beta + ??) = 1$ 則 左 $Max = \sqrt{2-2\cos\alpha}$

所以 $\sqrt{2-2\cos\alpha} \geq \sqrt{2} - \cos\alpha$ (註 $\sqrt{2} - \cos\alpha > 0$)

兩邊平方 $2 - 2\cos\alpha \geq 2 - 2\sqrt{2}\cos\alpha + \cos^2\alpha$

$\Rightarrow \cos^2\alpha + (2-2\sqrt{2})\cos\alpha \leq 0$

$\Rightarrow \cos\alpha[\cos\alpha - (2\sqrt{2}-2)] \leq 0$

$\Rightarrow 0 \leq \cos\alpha \leq 2\sqrt{2} - 2$

$\therefore$ Max $2\sqrt{2} - 2$

α 為一實數 有一點 $P(\cos\beta, \sin\beta)$ 且滿足 $\cos\alpha + \cos\beta - \cos(\alpha+\beta) = 2\cos\frac{\pi}{4}$

試求 $\cos\alpha$ 的最大值

$+\cos\beta - \cos\alpha\cos\beta + \sin\alpha\sin\beta = \sqrt{2}$

$\sin\alpha\sin\beta)^2 = (\sqrt{2} - \cos\alpha - \cos\beta + \cos\alpha\cos\beta)^2$

$=$

$\sin^2\alpha\sin^2\beta$

$=$

$(1-\cos^2\alpha)(1-\cos^2\beta) =$

$$\frac{4\sqrt{3}}{3} - \frac{\sqrt{3}}{2} + \frac{5\sqrt{3}}{6}$$

$$= \frac{8\sqrt{3} - 3\sqrt{3} + 5\sqrt{3}}{6}$$

$$= \frac{10\sqrt{3}}{6}$$

$$\frac{4\sqrt{3}}{3} - \frac{\sqrt{3}}{2} + \frac{5\sqrt{3}}{6}$$

$$\frac{4\sqrt{3}}{\sqrt{3}\times\sqrt{3}} - \frac{\sqrt{3}}{1} + \frac{5\sqrt{3}}{2\sqrt{3}\times\sqrt{3}}$$

$$\sqrt{32} = 3\times\sqrt{32} = 3\times 4\times\sqrt{2} = 12\sqrt{2}$$

$$\frac{5}{2}\times\frac{5}{12}\div\frac{3}{2}$$

$$\frac{5}{12}\times\frac{5}{12}\times\frac{2}{3}$$

$$= \frac{25}{216}$$

$$\sqrt{12} - \sqrt{75} + \sqrt{27} - \sqrt{48}$$

$$= 2\sqrt{3} - 5\sqrt{3} + 3\sqrt{3} - 4\sqrt{3}$$

$$= -4\sqrt{3}$$

$$2.222\ldots = 2.\dot{2}$$

$$\frac{3}{100} = 0.03$$

$$\frac{3}{16} = 0.1875$$

$$\frac{3}{8} = 0.375$$

$$\frac{5}{8} = 0.625$$

$$\frac{7}{8} = 0.875$$

$$\frac{1}{4} = 0.25$$

$$\frac{3}{4} = 0.75$$

$$\frac{1}{2} = 0.5$$

$$\frac{1}{3} = 0.\dot{3}$$

$$\frac{2}{3} = 0.\dot{6}$$

$$\frac{1}{6} = 0.1\dot{6}$$

$$\frac{5}{6} = 0.8\dot{3}$$

$$\frac{1}{9} = 0.\dot{1}$$

$$\frac{2}{9} = 0.\dot{2}$$

$$\frac{4}{9} = 0.\dot{4}$$

$$\frac{5}{9} = 0.\dot{5}$$

$$\frac{7}{9} = 0.\dot{7}$$

$$\frac{8}{9} = 0.\dot{8}$$

$$\frac{1}{11} = 0.\dot{0}\dot{9}$$

$$\frac{2}{11} = 0.\dot{1}\dot{8}$$

$$\frac{3}{11} = 0.\dot{2}\dot{7}$$

$$\frac{4}{11} = 0.\dot{3}\dot{6}$$

$$\frac{5}{11} = 0.\dot{4}\dot{5}$$

$$\frac{6}{11} = 0.\dot{5}\dot{4}$$

$$\frac{7}{11} = 0.\dot{6}\dot{3}$$

$$\frac{8}{11} = 0.\dot{7}\dot{2}$$

$$\frac{9}{11} = 0.\dot{8}\dot{1}$$

$$\frac{10}{11} = 0.\dot{9}\dot{0}$$

$$\frac{1}{12} = 0.08\dot{3}$$

$$\frac{5}{12} = 0.41\dot{6}$$

$$\frac{7}{12} = 0.58\dot{3}$$

$$\frac{11}{12} = 0.91\dot{6}$$

$$\frac{1}{13} = 0.\dot{0}7692\dot{3}$$

$$\frac{2}{13} = 0.\dot{1}5384\dot{6}$$

$$\frac{3}{13} = 0.\dot{2}3076\dot{9}$$

$$\frac{4}{13} = 0.\dot{3}0769\dot{2}$$

$$\frac{5}{13} = 0.\dot{3}8461\dot{5}$$

$$\frac{6}{13} = 0.\dot{4}6153\dot{8}$$

$$\frac{7}{13} = 0.\dot{5}3846\dot{1}$$

$$\frac{8}{13} = 0.\dot{6}1538\dot{4}$$

$$\frac{9}{13} = 0.\dot{6}9230\dot{7}$$

$$\frac{10}{13} = 0.\dot{7}6923\dot{0}$$

$$\frac{11}{13} = 0.\dot{8}4615\dot{3}$$

$$\frac{12}{13} = 0.\dot{9}2307\dot{6}$$

$$\frac{1}{14} = 0.0\dot{7}1428\dot{5}$$

$$\frac{3}{14} = 0.2\dot{1}4285\dot{7}$$

$$\frac{5}{14} = 0.3\dot{5}7142\dot{8}$$

$$\frac{9}{14} = 0.6\dot{4}2857\dot{1}$$

$$\frac{11}{14} = 0.7\dot{8}5714\dot{2}$$

$$\frac{13}{14} = 0.9\dot{2}8571\dot{4}$$

$$\frac{1}{15} = 0.0\dot{6}$$

$$\frac{2}{15} = 0.1\dot{3}$$

$$\frac{4}{15} = 0.2\dot{6}$$

$$\frac{7}{15} = 0.4\dot{6}$$

$$\frac{8}{15} = 0.5\dot{3}$$

$$\frac{11}{15} = 0.7\dot{3}$$

$$\frac{13}{15} = 0.8\dot{6}$$

$$\frac{14}{15} = 0.9\dot{3}$$

$$\frac{1}{16} = 0.0625$$

$$\frac{3}{16} = 0.1875$$

$$\frac{5}{16} = 0.3125$$

$$\frac{7}{16} = 0.4375$$

$$\frac{9}{16} = 0.5625$$

$$\frac{11}{16} = 0.6875$$

$$\frac{13}{16} = 0.8125$$

$$\frac{15}{16} = 0.9375$$

$$\frac{1}{17} = 0.\dot{0}5882352\dot{9}$$

$$\frac{2}{17} = 0.\dot{1}1764705\dot{8}$$

$$\frac{3}{17} = 0.\dot{1}7647058\dot{8}$$

$$\frac{4}{17} = 0.\dot{2}3529411\dot{7}$$

$$\frac{5}{17} = 0.\dot{2}9411764\dot{7}$$

$$\frac{6}{17} = 0.\dot{3}5294117\dot{6}$$

$$\frac{7}{17} = 0.\dot{4}1176470\dot{6}$$

$$\frac{8}{17} = 0.\dot{4}7058823\dot{5}$$

$$\frac{9}{17} = 0.\dot{5}2941176\dot{5}$$

$$\frac{10}{17} = 0.\dot{5}8823529\dot{4}$$

$$\frac{11}{17} = 0.\dot{6}4705882\dot{4}$$

$$\frac{12}{17} = 0.\dot{7}0588235\dot{3}$$

$$\frac{13}{17} = 0.\dot{7}6470588\dot{3}$$

$$\frac{14}{17} = 0.\dot{8}2352941\dot{2}$$

$$\frac{15}{17} = 0.\dot{8}8235294\dot{1}$$

$$\frac{16}{17} = 0.\dot{9}4117647\dot{1}$$

$$\frac{1}{18} = 0.0\dot{5}$$

$$\frac{5}{18} = 0.2\dot{7}$$

$$\frac{7}{18} = 0.3\dot{8}$$

$$\frac{11}{18} = 0.6\dot{1}$$

$$\frac{13}{18} = 0.7\dot{2}$$

$$\frac{17}{18} = 0.9\dot{4}$$

$$\frac{1}{19} = 0.\dot{0}52631578\dot{9}$$

$$\frac{2}{19} = 0.\dot{1}05263157\dot{8}$$

$$\frac{3}{19} = 0.\dot{1}57894736\dot{8}$$

$$\frac{4}{19} = 0.\dot{2}10526315\dot{7}$$

$$\frac{5}{19} = 0.\dot{2}63157894\dot{7}$$

$$\frac{6}{19} = 0.\dot{3}15789473\dot{6}$$

$$\frac{7}{19} = 0.\dot{3}68421052\dot{6}$$

$$\frac{8}{19} = 0.\dot{4}21052631\dot{5}$$

$$\frac{9}{19} = 0.\dot{4}73684210\dot{5}$$

$$\frac{10}{19} = 0.\dot{5}26315789\dot{4}$$

$$\frac{11}{19} = 0.\dot{5}78947368\dot{4}$$

$$\frac{12}{19} = 0.\dot{6}31578947\dot{3}$$

$$\frac{13}{19} = 0.\dot{6}84210526\dot{3}$$

$$\frac{14}{19} = 0.\dot{7}36842105\dot{2}$$

$$\frac{15}{19} = 0.\dot{7}89473684\dot{2}$$

$$\frac{16}{19} = 0.\dot{8}42105263\dot{1}$$

$$\frac{17}{19} = 0.\dot{8}94736842\dot{1}$$

$$\frac{18}{19} = 0.\dot{9}47368421\dot{0}$$

$$\frac{1}{20} = 0.05$$

$$\frac{3}{20} = 0.15$$

$$\frac{7}{20} = 0.35$$

$$\frac{9}{20} = 0.45$$

$$\frac{11}{20} = 0.55$$

$$\frac{13}{20} = 0.65$$

$$\frac{17}{20} = 0.85$$

$$\frac{19}{20} = 0.95$$

$$\frac{1}{21} = 0.\dot{0}47619\dot{0}$$

$$\frac{2}{21} = 0.\dot{0}95238\dot{1}$$

$$\frac{4}{21} = 0.\dot{1}90476\dot{2}$$

$$\frac{5}{21} = 0.\dot{2}38095\dot{2}$$

$$\frac{8}{21} = 0.\dot{3}80952\dot{3}$$

$$\frac{10}{21} = 0.\dot{4}76190\dot{4}$$

$$\frac{11}{21} = 0.\dot{5}23809\dot{5}$$

$$\frac{13}{21} = 0.\dot{6}19047\dot{6}$$

$$\frac{16}{21} = 0.\dot{7}61904\dot{7}$$

$$\frac{17}{21} = 0.\dot{8}09523\dot{8}$$

$$\frac{19}{21} = 0.\dot{9}04761\dot{9}$$

$$\frac{20}{21} = 0.\dot{9}52380\dot{9}$$

$$\frac{1}{22} = 0.0\dot{4}\dot{5}$$

$$\frac{3}{22} = 0.1\dot{3}\dot{6}$$

$$\frac{5}{22} = 0.2\dot{2}\dot{7}$$

$$\frac{7}{22} = 0.3\dot{1}\dot{8}$$

$$\frac{9}{22} = 0.4\dot{0}\dot{9}$$

$$\frac{13}{22} = 0.5\dot{9}\dot{0}$$

$$\frac{15}{22} = 0.6\dot{8}\dot{1}$$

$$\frac{17}{22} = 0.7\dot{7}\dot{2}$$

$$\frac{19}{22} = 0.8\dot{6}\dot{3}$$

$$\frac{21}{22} = 0.9\dot{5}\dot{4}$$

$$\frac{1}{23} = 0.\dot{0}43478260869565\dot{2}$$

$$\frac{2}{23} = 0.\dot{0}86956521739130\dot{4}$$

$$\frac{3}{23} = 0.\dot{1}30434782608695\dot{6}$$

$$\frac{4}{23} = 0.\dot{1}73913043478260\dot{8}$$

$$\frac{5}{23} = 0.\dot{2}17391304347826\dot{0}$$

$$\frac{6}{23} = 0.\dot{2}60869565217391\dot{2}$$

$$\frac{7}{23} = 0.\dot{3}04347826086956\dot{5}$$

$$\frac{8}{23} = 0.\dot{3}47826086956521\dot{7}$$

$$\frac{9}{23} = 0.\dot{3}91304347826086\dot{9}$$

$$\frac{10}{23} = 0.\dot{4}34782608695652\dot{1}$$

$$\frac{11}{23} = 0.\dot{4}78260869565217\dot{3}$$

$$\frac{12}{23} = 0.\dot{5}21739130434782\dot{6}$$

$$\frac{13}{23} = 0.\dot{5}65217391304347\dot{8}$$

$$\frac{14}{23} = 0.\dot{6}08695652173913\dot{0}$$

$$\frac{15}{23} = 0.\dot{6}52173913043478\dot{2}$$

$$\frac{16}{23} = 0.\dot{6}95652173913043\dot{4}$$

$$\frac{17}{23} = 0.\dot{7}39130434782608\dot{7}$$

$$\frac{18}{23} = 0.\dot{7}82608695652173\dot{9}$$

$$\frac{19}{23} = 0.\dot{8}26086956521739\dot{1}$$

$$\frac{20}{23} = 0.\dot{8}69565217391304\dot{3}$$

$$\frac{21}{23} = 0.\dot{9}13043478260869\dot{5}$$

$$\frac{22}{23} = 0.\dot{9}56521739130434\dot{7}$$

$$\frac{1}{24} = 0.041\dot{6}$$

$$\frac{5}{24} = 0.208\dot{3}$$

$$\frac{7}{24} = 0.291\dot{6}$$

$$\frac{11}{24} = 0.458\dot{3}$$

$$\frac{13}{24} = 0.541\dot{6}$$

$$\frac{17}{24} = 0.708\dot{3}$$

$$\frac{19}{24} = 0.791\dot{6}$$

$$\frac{23}{24} = 0.958\dot{3}$$

$$\frac{1}{25} = 0.04$$

$$\frac{3}{25} = 0.12$$

$$\frac{4}{25} = 0.16$$

$$\frac{6}{25} = 0.24$$

$$\frac{7}{25} = 0.28$$

$$\frac{8}{25} = 0.32$$

$$\frac{9}{25} = 0.36$$

$$\frac{11}{25} = 0.44$$

$$\frac{12}{25} = 0.48$$

$$\frac{13}{25} = 0.52$$

$$\frac{14}{25} = 0.56$$

$$\frac{16}{25} = 0.64$$

$$\frac{17}{25} = 0.68$$

$$\frac{18}{25} = 0.72$$

$$\frac{19}{25} = 0.76$$

$$\frac{21}{25} = 0.84$$

$$\frac{22}{25} = 0.88$$

$$\frac{23}{25} = 0.92$$

$$\frac{24}{25} = 0.96$$

$$\frac{1}{26} = 0.0\dot{3}84615\dot{3}$$

$$\frac{3}{26} = 0.1\dot{1}53846\dot{1}$$

$$\frac{5}{26} = 0.1\dot{9}23076\dot{9}$$

$$\frac{7}{26} = 0.2\dot{6}92307\dot{6}$$

$$\frac{9}{26} = 0.3\dot{4}61538\dot{4}$$

$$\frac{11}{26} = 0.4\dot{2}30769\dot{2}$$

$$\frac{15}{26} = 0.5\dot{7}69230\dot{7}$$

$$\frac{17}{26} = 0.6\dot{5}38461\dot{5}$$

$$\frac{19}{26} = 0.7\dot{3}07692\dot{3}$$

$$\frac{21}{26} = 0.8\dot{0}76923\dot{0}$$

$$\frac{23}{26} = 0.8\dot{8}46153\dot{8}$$

$$\frac{25}{26} = 0.9\dot{6}15384\dot{6}$$

$$\frac{1}{27} = 0.\dot{0}37\dot{0}$$

$$\frac{2}{27} = 0.\dot{0}74\dot{1}$$

$$\frac{4}{27} = 0.\dot{1}48\dot{1}$$

$$\frac{5}{27} = 0.\dot{1}85\dot{1}$$

$$\frac{7}{27} = 0.\dot{2}59\dot{2}$$

$$\frac{8}{27} = 0.\dot{2}96\dot{2}$$

$$\frac{10}{27} = 0.\dot{3}70\dot{3}$$

$$\frac{11}{27} = 0.\dot{4}07\dot{4}$$

$$\frac{13}{27} = 0.\dot{4}81\dot{4}$$

$$\frac{14}{27} = 0.\dot{5}18\dot{5}$$

$$\frac{16}{27} = 0.\dot{5}92\dot{5}$$

$$\frac{17}{27} = 0.\dot{6}29\dot{6}$$

$$\frac{19}{27} = 0.\dot{7}03\dot{7}$$

$$\frac{20}{27} = 0.\dot{7}40\dot{7}$$

$$\frac{22}{27} = 0.\dot{8}14\dot{8}$$

$$\frac{23}{27} = 0.\dot{8}51\dot{8}$$

$$\frac{25}{27} = 0.\dot{9}25\dot{9}$$

$$\frac{26}{27} = 0.\dot{9}62\dot{9}$$

$$\frac{1}{28} = 0.03\dot{5}71\dot{4}$$

$$\frac{3}{28} = 0.10\dot{7}14\dot{2}$$

$$\frac{5}{28} = 0.17\dot{8}57\dot{1}$$

$$\frac{9}{28} = 0.32\dot{1}42\dot{8}$$

$$\frac{11}{28} = 0.39\dot{2}85\dot{7}$$

$$\frac{13}{28} = 0.46\dot{4}28\dot{5}$$

$$\frac{15}{28} = 0.53\dot{5}71\dot{4}$$

$$\frac{17}{28} = 0.60\dot{7}14\dot{2}$$

$$\frac{19}{28} = 0.67\dot{8}57\dot{1}$$

$$\frac{23}{28} = 0.82\dot{1}42\dot{8}$$

$$\frac{25}{28} = 0.89\dot{2}85\dot{7}$$

$$\frac{27}{28} = 0.96\dot{4}28\dot{5}$$

$$\frac{1}{29} = 0.\dot{0}34482758620689\dot{6}55172413793103\dot{4}$$

$$\frac{2}{29} = 0.\dot{0}68965517241379\dot{3}10344827586206\dot{9}$$

$$\frac{3}{29} = 0.\dot{1}03448275862068\dot{9}65517241379310\dot{3}$$

$$\frac{4}{29} = 0.\dot{1}37931034482758\dot{6}20689655172413\dot{7}$$

$$\frac{5}{29} = 0.\dot{1}72413793103448\dot{2}75862068965517\dot{2}$$

$$\frac{6}{29} = 0.\dot{2}06896551724137\dot{9}31034482758620\dot{6}$$

$$\frac{7}{29} = 0.\dot{2}41379310344827\dot{5}86206896551724\dot{1}$$

$$\frac{8}{29} = 0.\dot{2}75862068965517\dot{2}41379310344827\dot{5}$$

$$\frac{9}{29} = 0.\dot{3}10344827586206\dot{8}96551724137931\dot{0}$$

$$\frac{10}{29} = 0.\dot{3}44827586206896\dot{5}51724137931034\dot{4}$$

$$\frac{11}{29} = 0.\dot{3}79310344827586\dot{2}06896551724137\dot{9}$$

$$\frac{12}{29} = 0.\dot{4}13793103448275\dot{8}62068965517241\dot{3}$$

$$\frac{13}{29} = 0.\dot{4}48275862068965\dot{5}17241379310344\dot{8}$$

$$\frac{14}{29} = 0.\dot{4}82758620689655\dot{1}72413793103448\dot{2}$$

$$\frac{15}{29} = 0.\dot{5}17241379310344\dot{8}27586206896551\dot{7}$$

$$\frac{16}{29} = 0.\dot{5}51724137931034\dot{4}82758620689655\dot{1}$$

$$\frac{17}{29} = 0.\dot{5}86206896551724\dot{1}37931034482758\dot{6}$$

$$\frac{18}{29} = 0.\dot{6}20689655172413\dot{7}93103448275862\dot{0}$$

$$\frac{19}{29} = 0.\dot{6}55172413793103\dot{4}48275862068965\dot{5}$$

$$\frac{20}{29} = 0.\dot{6}89655172413793\dot{1}03448275862068\dot{9}$$

$$\frac{21}{29} = 0.\dot{7}24137931034482\dot{7}58620689655172\dot{4}$$

$$\frac{22}{29} = 0.\dot{7}58620689655172\dot{4}13793103448275\dot{8}$$

$$\frac{23}{29} = 0.\dot{7}93103448275862\dot{0}68965517241379\dot{3}$$

$$\frac{24}{29} = 0.\dot{8}27586206896551\dot{7}24137931034482\dot{7}$$

$$\frac{25}{29} = 0.\dot{8}62068965517241\dot{3}79310344827586\dot{2}$$

$$\frac{26}{29} = 0.\dot{8}96551724137931\dot{0}34482758620689\dot{6}$$

$$\frac{27}{29} = 0.\dot{9}31034482758620\dot{6}89655172413793\dot{1}$$

$$\frac{28}{29} = 0.\dot{9}65517241379310\dot{3}44827586206896\dot{5}$$

$$\frac{1}{30} = 0.0\dot{3}$$

$$\frac{7}{30} = 0.2\dot{3}$$

$$\frac{11}{30} = 0.3\dot{6}$$

$$\frac{13}{30} = 0.4\dot{3}$$

$$\frac{17}{30} = 0.5\dot{6}$$

$$\frac{19}{30} = 0.6\dot{3}$$

$$\frac{23}{30} = 0.7\dot{6}$$

$$\frac{29}{30} = 0.9\dot{6}$$

$$\frac{1}{31} = 0.\dot{0}32258064516129\dot{0}$$

$$\frac{2}{31} = 0.\dot{0}64516129032258\dot{0}$$

$$\frac{3}{31} = 0.\dot{0}96774193548387\dot{0}$$

$$\frac{4}{31} = 0.\dot{1}29032258064516\dot{0}$$

$$\frac{5}{31} = 0.\dot{1}61290322580645\dot{0}$$

$$\frac{6}{31} = 0.\dot{1}93548387096774\dot{0}$$

$$\frac{7}{31} = 0.\dot{2}25806451612903\dot{0}$$

$$\frac{8}{31} = 0.\dot{2}58064516129032\dot{0}$$

$$\frac{9}{31} = 0.\dot{2}90322580645161\dot{0}$$

$$\frac{10}{31} = 0.\dot{3}22580645161290\dot{0}$$

$$\frac{11}{31} = 0.\dot{3}54838709677419\dot{0}$$

$$\frac{12}{31} = 0.\dot{3}87096774193548\dot{0}$$

$$\frac{13}{31} = 0.\dot{4}19354838709677\dot{0}$$

$$\frac{14}{31} = 0.\dot{4}51612903225806\dot{0}$$

$$\frac{15}{31} = 0.\dot{4}83870967741935\dot{0}$$

$$\frac{16}{31} = 0.\dot{5}16129032258064\dot{0}$$

$$\frac{17}{31} = 0.\dot{5}48387096774193\dot{0}$$

$$\frac{18}{31} = 0.\dot{5}80645161290322\dot{0}$$

$$\frac{19}{31} = 0.\dot{6}12903225806451\dot{0}$$

$$\frac{20}{31} = 0.\dot{6}45161290322580\dot{0}$$

$$\frac{21}{31} = 0.\dot{6}77419354838709\dot{0}$$

$$\frac{22}{31} = 0.\dot{7}09677419354838\dot{0}$$

$$\frac{23}{31} = 0.\dot{7}41935483870967\dot{0}$$

$$\frac{24}{31} = 0.\dot{7}74193548387096\dot{0}$$

$$\frac{25}{31} = 0.\dot{8}06451612903225\dot{0}$$

$$\frac{26}{31} = 0.\dot{8}38709677419354\dot{0}$$

$$\frac{27}{31} = 0.\dot{8}70967741935483\dot{0}$$

$$\frac{28}{31} = 0.\dot{9}03225806451612\dot{0}$$

$$\frac{29}{31} = 0.\dot{9}35483870967741\dot{0}$$

$$\frac{30}{31} = 0.\dot{9}67741935483870\dot{0}$$

$$\frac{1}{32} = 0.03125$$

$$\frac{3}{32} = 0.09375$$

$$\frac{5}{32} = 0.15625$$

$$\frac{7}{32} = 0.21875$$

$$\frac{9}{32} = 0.28125$$

$$\frac{11}{32} = 0.34375$$

$$\frac{13}{32} = 0.40625$$

$$\frac{15}{32} = 0.46875$$

$$\frac{17}{32} = 0.53125$$

$$\frac{19}{32} = 0.59375$$

$$\frac{21}{32} = 0.65625$$

$$\frac{23}{32} = 0.71875$$

$$\frac{25}{32} = 0.78125$$

$$\frac{27}{32} = 0.84375$$

$$\frac{29}{32} = 0.90625$$

$$\frac{31}{32} = 0.96875$$

1A1

$$\frac{8(\sqrt{11}+3)}{(\sqrt{11}-3)(\sqrt{11}+3)} = \frac{4 \cdot 2(\sqrt{11}+3)}{11-9}$$

$$= 4(\sqrt{11}+3)$$

$$= 4\sqrt{11} + 12.$$

$$\frac{\sqrt{16}-1}{3} = \frac{3}{3}.$$

$$\frac{\sqrt{4}-1+\sqrt{9}-\sqrt{4}+\sim\sqrt{16}-\sqrt{9}}{3}$$

$$\frac{(\sqrt{4}+1)\times(\sqrt{4}-1)}{(\sqrt{4}-1)} = \frac{4-1}{\sqrt{4}-1} = \frac{3}{2-1} = 3$$

$$(a+b)(a-b) = a^2-b^2 \quad \text{Co} \left| \frac{2}{1} \right.$$

$$x+y = 9t$$

$$y+z = 6t$$

$$z+x = 4t$$

$$2(x+y+z) = 19t$$

$$x+y+z = \frac{19t}{2}$$

$$z = \frac{19t}{2} - 9t = \frac{1}{2}t$$

$$y = \frac{19t}{2} - 4t = \frac{11}{2}t$$

$$x = \frac{19t}{2} - 6t = \frac{7}{2}t$$

$$\frac{x}{z} = \frac{\frac{7}{2}t}{\frac{1}{2}t} = 7$$

-15.

乙 = ( ).

甲+乙+丙 = ( )

丙 = ?

甲+乙+4丙 = [ ]

A ∪ (5)

B D 乙 -5

-10

$\cos(A-B) = \frac{4}{5}$

$\Rightarrow \cos^2 A\cos^2 B + \sin^2 A \sin^2 B = \frac{4}{5}$

$\Rightarrow (\sin A \sin B)^2 = (\frac{4}{5} - \cos A \cos B)^2$

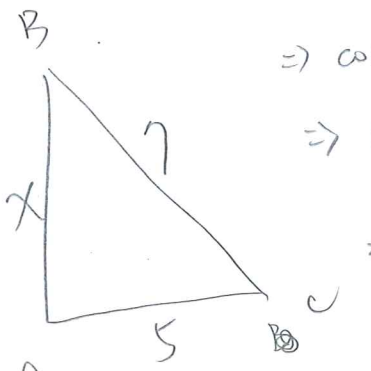
$\Rightarrow \sin^2 A \sin^2 B = \frac{16}{25} - \frac{8}{5}\cos A\cos B + \cos^2 A \cos^2 B$

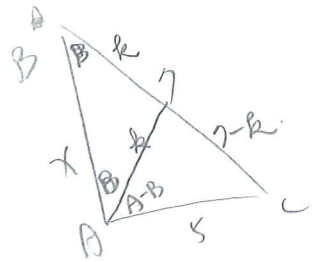
$\Rightarrow (1-\cos^2 A)(1-\cos^2 B) = \frac{16}{25} - \frac{8}{5}\cos A\cos B + \cos^2 A\cos^2 B$

$\Rightarrow 1 - \cos^2 A - \cos^2 B + \cos^2 A\cos^2 B = \frac{16}{25} - \frac{8}{5}\cos A\cos B + \cos^2 A\cos^2 B$

$\cos A = \frac{x^2 + 5^2 - 7^2}{2\times 5 \times x}$

$\cos B = \frac{x^2 + 7^2 - 5^2}{2\times x \times 7}$





11,3258
7,6793
3,6465

4,722
8,3986
13,1206

1,0945
0,378
0,7165

8 | 24,32
3,4

5,0236
4,9764
10,0000

11,3258
7,1793
65

12,307
6,5883
18,8953

9,3074
10,6926

16 | 16,48
1,5

9 | 18,81
2,1

8 | 72,80
9,10

2 | 28,42
14,21
2,3

3 | 18,24
6,8

9 | 18,27
2,3

4 | 72,80
8
18,2

6 | 18,24
3,4

7 | 35,63
5,9

13,582
12,691
30,273

6 | 36,54
3 | 6,9
2,3

4 | 56,20
14,5

4,722
8,3986
0,6

17,00

$+\cos\beta-\cos(\alpha+\beta)=\sqrt{2}$

$+\cos\beta-\cos\alpha\cos\beta+\sin\alpha\sin\beta=\sqrt{2}$ $\rightsquigarrow$ $(\cos\alpha+\cos\beta-\cos\alpha\cos\beta)^2=(\sqrt{2}-\sin\alpha\sin\beta)^2$

$\alpha(1-\cos\beta)+\sin\alpha\sin\beta=\sqrt{2}-\cos\beta$ $\Rightarrow$ $\cos^2\alpha+\cos^2\beta+2\cos\alpha\cos\beta$

$\cos\alpha(1-\cos\beta)+\sqrt{1-\cos^2\alpha}\sin\beta=\sqrt{2}-\cos\beta$

$\cos\alpha(1-\cos\beta)$ =

$\alpha+\cos\beta-\cos\alpha\cos\beta)(\cos\alpha+\cos\beta-\cos\alpha\cos\beta)$

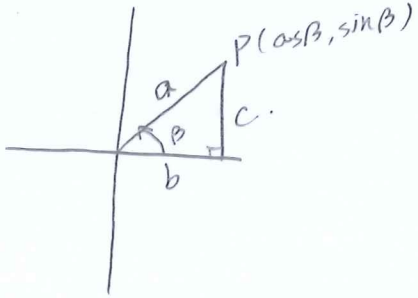
$\cos^2\alpha+\cos\alpha\cos\beta-\cos^2\alpha\cos\beta+\cos\alpha\cos\beta-\cos\alpha\cos^2\beta-\cos^2\alpha\cos\beta-\cos\alpha\cos^2\beta+\cos^2\alpha\cos^2\beta+\cos^2\beta$

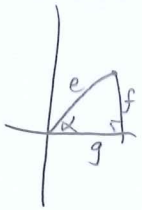
$\cos^2\alpha+2\cos\alpha\cos\beta-2\cos^2\alpha\cos\beta-2\cos\alpha\cos^2\beta+\cos^2\alpha\cos^2\beta$

~~$\cos\alpha + \cos\beta - \cos\alpha\cos\beta + \sin\alpha\sin\beta = \sqrt{2}$~~

~~$X = \cos\alpha + \cos\beta$~~

~~$X^2 = \cos^2\alpha + \cos^2\beta + 2\cos\alpha\cos\beta$~~

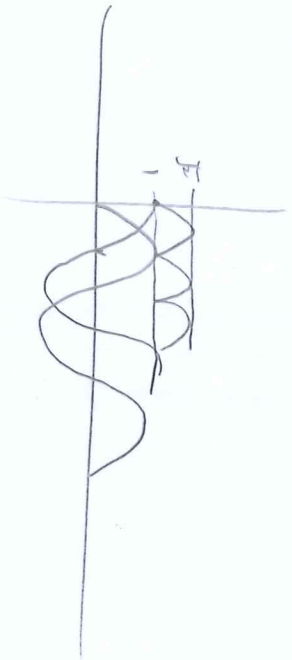




$\cos\alpha = \frac{g}{e}$

$\sin\alpha = \sqrt{\sin^2\alpha} = \sqrt{1-\cos^2\alpha}$

$|\sin x| + |\cos x|$



$\cos\alpha + \cos\beta - \cos\alpha\cos\beta + \sin\alpha\sin\beta = \sqrt{2}$

$\frac{g}{e} + \frac{b}{a} - \frac{g}{e} \cdot \frac{b}{a} + \frac{f}{e} \cdot \frac{c}{a} = \sqrt{2}$

$\frac{95.7}{15} \approx$ ~~6.30~~ 6.38.

$6.38 \times 2 = 12.76$

$6.38 \times 3 = 19.14$ ↓

$95.7 - 12.76. = (82.9\sim)$

$95.7 - 19.14 = 76.\sim$

$\lim_{n\to\infty} \frac{1}{\sqrt{n+1}+\sqrt{n}} = 0$

$= \frac{(\sqrt{n+1}-\sqrt{n})}{(\sqrt{n+1}+\sqrt{n})(\sqrt{n+1}-\sqrt{n})}$

$= \frac{\sqrt{n+1}-\sqrt{n}}{(n+1)-n}$

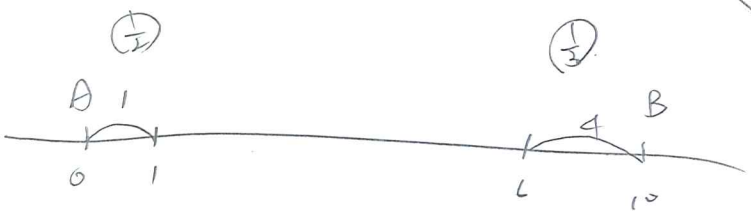
$= \sqrt{n+1}-\sqrt{n}$

$\lim_{n\to\infty} (\sqrt{n+1}-\sqrt{n}) = \infty - \infty$

$\frac{\text{분자}}{\text{분모}}$ 분모의 최고차항으로 나눈다.

$\frac{\text{분자}}{\text{분모}}$ 분자의 최고차항으로 나눈다.

$\frac{\text{분자}}{\text{분모}}$ 분자, 분모 $\sqrt{}$ 있으면 유리화



$C_n = C_{n-1} + \cdots$

$C_n = 1 + (\frac{1}{2}) + (\frac{1}{2})^2 + \sim + (\frac{1}{2})^{n-1} = \frac{1\times[1-(\frac{1}{2})^n]}{1-\frac{1}{2}} = 2[1-(\frac{1}{2})^n]$.

$b_n = 10 - 4 - \frac{4}{3} - \frac{4}{9} - \cdots - \frac{4}{3^{n-1}} = 10 - 4(1 + \frac{1}{3} + \sim \frac{1}{3^{n-1}})$

$= 10 - 4 \times \frac{1\times[1-(\frac{1}{3})^n]}{1-\frac{1}{3}} = 10 - 6[1-(\frac{1}{3})^n]$.

$\lim_{n\to\infty} C_n = \lim_{n\to\infty} \frac{2[1-(\frac{1}{2})^n] + 10 - 6[1-(\frac{1}{3})^n]}{2} = \frac{2+10-6}{2} = 3$.

$$f \circ g(x) = f(gx))$$

$$= f(5+4x-x^2)$$

$$= \sqrt{5+4x-x^2} = \sqrt{-(x-2)^2+9} \geq 3$$

$$5+4x-x^2 \geq 0$$

$$x^2-4x-5 \leq 0$$

$$-1 \leq x \leq 5$$

$$\frac{g(x)}{f(x)} = \frac{\sqrt{5+4x-x^2}}{5x} = \frac{0}{15} = 0.$$

$$x \to 5^+$$

$$0$$

$$[1.8] = 1.$$

$$\lim_{x \to -1^+} \frac{x+1}{x+1} = 1.$$

$$\lim_{x \to -1^-} \frac{x+1}{-(x+1)} = -1$$

$$\lim_{x \to 2^+} ([x]-x) = 2-2 = 0$$

$$\lim_{x \to 2^-} ([x]-x) = 2-1 = 1.$$

$$\lim_{x \to 0} \frac{x \cdot (x+2)}{x} = 2.$$

$$\frac{\sin\theta}{\sin\varphi} = \frac{\sin\theta\cos\varphi}{\cos\varphi} = \frac{\sin\theta\cos\varphi}{\sqrt{1-\sin^2\varphi}} = \frac{\sin\theta\cos\varphi}{\sqrt{1-\sin^2\theta\cos^2\varphi}}$$

$$\frac{\sin\theta}{\sin\varphi} = \frac{\sin\theta\cos\varphi}{\sqrt{1-\sin^2\theta\cos^2\varphi}} = \frac{\sin\theta\cos\varphi}{\sqrt{1-\sin^2\theta+\sin^2\theta\sin^2\varphi}}$$

$$\frac{\sin\theta}{\sin\varphi} = \frac{\sin\theta\cos\varphi}{\sqrt{\cos^2\theta+\sin^2\theta\sin^2\varphi}} = \frac{\sin\theta\cos\varphi}{\sqrt{\cos

$S_n = \frac{(a_1+a_n)\times n}{2}$ 等差数列求和

$S_n = \frac{b_1\times(1-q^n)}{1-q}$ 等比数列求和

$a_1=1$ $a_n=a_1+(n-1)d$

$S_n = \frac{(a_1+a_n)\times n}{2}$

$a_n = a_1 + (n-1)d$ ⇒ $98 = (n-1)d$ —②

$S_n = \frac{a_1(1-q^n)}{1-q} = \frac{a_1-a_1q^n}{1-q}$

$a_1=1$ $a_n=a_1+(n-1)d$ ⇒ $a_n = 1+(n-1)d$

$q = \frac{a_n}{a_{n-1}}$ ⇒ $\frac{n-1}{n-1} = 10$ ⇒ $10(n-1) = n-1$

$n=2$ ⇒ $n=11$ ⇒ $n=1$

$n=3$ ⇒ $n=4$ ⇒ $n=1$

$a_1=1$, $a_n=98$, $d=?$

$98 = (n-1)d$ —②

$S_n = \frac{a_1(1-q^n)}{1-q} = \frac{a_1-a_1q^n}{1-q}$

$S_n = \frac{1-98q}{1-q}$

$S_n = \frac{98q-1}{q-1}$

$S_n = \frac{a_1(1-q^n)}{1-q} = \frac{a_1-a_1q^n}{1-q}$

$a_1=3$

$a_n = 384$

$S_n = 765$

$S_n = \frac{a_1(1-q^n)}{1-q} = \frac{a_1-a_1q^n}{1-q}$

$S_n = \frac{3-384q}{1-q}$

$765 = \frac{3-384q}{1-q}$

$765(1-q) = 3-384q$

$765-765q = 3-384q$

$765-3 = 765q-384q$

$762 = 381q$

$q = 2$

$a_n = a_1q^{n-1}$

$a_n = 3\times 2^{n-1}$

$S_n = \frac{a_1(1-q^n)}{1-q}$

$S_n = \frac{3(1-2^n)}{1-2}$

$S_n = 3(2^n-1)$

$$\frac{-24\sqrt{5}+6\sqrt{5}}{-6} = \frac{(-8\sqrt{3}+\sqrt{12})\times\sqrt{6}}{-\sqrt{6}\times\sqrt{6}}$$

$$-8\sqrt{3} + \sqrt{12} = -6\sqrt{3}$$

$$\frac{4\sqrt{3}\times\sqrt{6}}{2\sqrt{6}\times\sqrt{6}} = \frac{\sqrt{18}}{6} = \frac{3\sqrt{2}}{6}$$

$$= -10\sqrt{2}$$

$$7\sqrt{6} + 18 - 6\sqrt{2}$$

$$\sqrt{3}\times\sqrt{6} = \sqrt{18} = 3\sqrt{2}$$

$$-8\sqrt{18}$$

$$\sqrt{12} = 6\sqrt{2}$$

$$\frac{15\times\sqrt{3}}{\sqrt{4}\times\sqrt{3}} \times \frac{\sqrt{4}}{\sqrt{3}}$$

$$\frac{15\times\sqrt{3}}{2\times\sqrt{3}} \times \frac{2}{\sqrt{3}}$$

$$\frac{15\times\sqrt{3}\times 2}{2\times\sqrt{3}\times\sqrt{3}}$$

$$\frac{30\sqrt{3}}{6\times 3} = \frac{30\sqrt{3}}{18} = \frac{5\sqrt{3}}{3}$$

$$\frac{5\sqrt{3}}{3} \times \frac{\sqrt{3}}{\sqrt{3}}$$

$$\frac{5\times 3}{3\times\sqrt{3}} = \frac{15}{3\sqrt{3}}$$

$$\frac{15}{3\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$$

$$\frac{15\sqrt{3}}{3\times 3} = \frac{15\sqrt{3}}{9} = \frac{5\sqrt{3}}{3}$$

$$\frac{5\sqrt{3}}{3} \times \frac{\sqrt{3}}{\sqrt{3}}$$

$$\frac{5\times 3}{3\times\sqrt{3}} = \frac{15}{3\sqrt{3}}$$

$$\frac{15}{3\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$$

$$\frac{15\sqrt{3}}{3\times 3} = \frac{15\sqrt{3}}{9} = \frac{5\sqrt{3}}{3}$$

